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Computational HW#5
4/1/26

1a. Analytically compute the Fourier Transform $\mathcal{F}$ of

$$f(x) = e^{-50(x-1/2)^2}$$

Use the convention $\int_{-\infty}^{\infty} f(x)e^{-ikx}dx = \hat{f}(k)$. For reference, the inverse Fourier transform with this convention is $\frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(k)e^{ikx}dk = f(x)$.

Hint:

$$\int_{-\infty}^{\infty} e^{-ax^2+bx}dx = e^{\frac{b^2}{4a}}\sqrt{\frac{\pi}{a}}$$

$$\hat{f}(k) = \int_{-\infty}^{\infty} e^{-50(x-1/2)^2} e^{-ikx} dx$$

$$= \int_{-\infty}^{\infty} e^{-50(x-1/2)^2 - ikx} dx$$

$$= \int_{-\infty}^{\infty} e^{-50x^2+50x-12.5-ikx} dx$$

$$= \int_{-\infty}^{\infty} e^{-50x^2+x(50-ik)-12.5} dx$$

$$= e^{-12.5}\int_{-\infty}^{\infty} e^{-50x^2+x(50-ik)}$$

$$= e^{\frac{(50-ik)^2}{4(50)}}\sqrt{\frac{\pi}{50}}\, e^{-12.5}$$

$$= e^{\frac{2500}{200} - \frac{100ik}{200} - \frac{k^2}{200} - 12.5}\sqrt{\frac{\pi}{50}}$$

$$= e^{12.5 - 1/2ik - k^2/200 - 12.5}\sqrt{\frac{\pi}{50}}$$

$$\boxed{\hat{f}(k) = \sqrt{\frac{\pi}{50}}\, e^{-1/2ik - k^2/200}}$$

$$-50(x-1/2)^2 = -50(x-1/2)(x-1/2)$$
$$= -50(x^2 - 1/2x - 1/2x + 1/4)$$
$$= -50(x^2 - x + 1/4)$$
$$= -50x^2 + 50x - 12.5$$

In Gaussian
$a = 50$, $b = (50-ik)$

$$(50-ik)^2 = (50-ik)(50-ik)$$
$$= 2500 - 50ik - 50ik + i^2k^2$$
$$= 2500 - 100ik - k^2$$

$i^2 = -1$